

EE517 Analog VLSI Lab

Experiment 6

Layout Designing and Analysis of Two-Stage Operational
Amplifier



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1 Objective

The objective of this experiment is:

- To design layout of two-stage Miller compensated operational amplifier
- Perform DRC verification
- Perform LVS verification
- Perform parasitic extraction (PEX)
- Analyze performance degradation due to layout parasitics
- Compare pre-layout and post-layout results
- Compare performance of opamp
 - without null resistor
 - with null resistor

2 Theory

In analog integrated circuit design, schematic-level simulations provide an ideal representation of circuit performance by considering only the intended devices and components. However, during physical implementation of the circuit in layout, several additional effects are introduced due to fabrication constraints, interconnect routing, and device placement. These effects are collectively referred to as layout-dependent parasitic effects.

Layout design converts the schematic representation of a circuit into a geometrical structure consisting of diffusion regions, polysilicon gates, metal interconnect layers, contacts, and vias that can be fabricated using a specific CMOS technology process (TSMC 180 nm in this experiment). During this transformation, the electrical behavior of the circuit is influenced by unavoidable parasitic elements associated with the physical structure.

The major sources of performance variation introduced during layout implementation include:

- **Parasitic Capacitances**

Additional capacitances appear between metal layers, diffusion regions, polysilicon gates, and substrate. These parasitic capacitances increase node loading and introduce additional poles in the frequency response of the operational amplifier, which may reduce bandwidth and phase margin.

- **Parasitic Resistances**

Metal routing and contacts introduce distributed resistance along signal paths. These resistances increase signal delay, reduce gain at high frequencies, and degrade the slew rate performance of the amplifier.

- **Routing Asymmetry Effects**

Unequal routing lengths between matched devices such as differential pair transistors introduce mismatch in parasitic resistance and capacitance. This degrades differential gain accuracy and reduces the common-mode rejection ratio (CMRR).

- **Device Matching Limitations**

Process gradients across the chip may cause variations in threshold voltage, oxide thickness, and mobility. Improper placement of matched devices increases input offset voltage and reduces amplifier precision.

- **Long Interconnect Effects**

Long routing paths increase RC delay and introduce additional loading at internal nodes of the amplifier. This directly affects unity gain bandwidth (UGF), settling time, and stability of the two-stage operational amplifier.

Because schematic simulations do not include these layout-dependent parasitic effects, post-layout verification becomes essential to evaluate the realistic performance of the operational amplifier. Therefore, after completing the layout design, a sequence of verification steps is performed to ensure correctness and reliability of the physical implementation.

The key verification stages involved in post-layout analysis are:

- Design Rule Check (DRC)
- Layout Versus Schematic (LVS)
- Parasitic Extraction (PEX)
- Post-layout simulation using extracted netlist

These verification steps ensure that the implemented layout is fabrication-compliant, electrically equivalent to the schematic, and accurately represents the practical performance of the designed two-stage operational amplifier.

2.1 Design Rule Check (DRC)

Design Rule Check (DRC) is an essential physical verification step performed after completing the layout of an integrated circuit. The objective of DRC is to ensure that the layout strictly follows the fabrication constraints specified by the semiconductor foundry for the selected technology node (TSMC 180 nm in this experiment). These rules guarantee manufacturability, reliability, and yield of the fabricated chip.

During layout implementation, geometrical shapes such as diffusion regions, polysilicon gates, metal layers, contacts, and vias must satisfy minimum spacing and enclosure requirements. Violation of these constraints may lead to fabrication defects such as open circuits, short circuits, excessive leakage currents, or device mismatch.

Typical DRC rules verified during this experiment include:

- **Minimum Width Rule:** Ensures that the width of diffusion, polysilicon, metal layers, and interconnects is not smaller than the minimum allowed value. Narrow wires may break during fabrication and increase resistance significantly.

- **Minimum Spacing Rule:** Ensures that adequate spacing exists between adjacent layout geometries belonging to the same or different layers. Insufficient spacing may cause short circuits after fabrication due to lithographic limitations.
- **Overlap and Enclosure Rule:** Ensures that contacts and vias are properly enclosed by surrounding layers such as metal or diffusion regions. Proper enclosure guarantees reliable electrical connectivity.
- **Contact and Via Rules:** Ensures correct placement and spacing of contacts between diffusion–metal and metal–metal layers. Improper contact placement can introduce large parasitic resistance or connectivity failures.
- **Well and Implant Rules:** Ensure correct placement of NMOS and PMOS devices inside their respective wells. These rules are critical to prevent latch-up and leakage paths.
- **Antenna Rules:** Ensure that long metal routing connected to transistor gates does not accumulate excessive charge during fabrication plasma processes. Violation of antenna rules can permanently damage thin gate oxide layers.

After running the DRC verification in Cadence Virtuoso using Calibre, no rule violations were detected. Therefore, the layout is considered **DRC clean** and suitable for further verification steps such as LVS and parasitic extraction.

2.2 Layout Versus Schematic (LVS)

Layout Versus Schematic (LVS) verification is a critical step in the physical design flow of integrated circuits. The purpose of LVS is to ensure that the electrical connectivity and device parameters in the layout exactly match those defined in the schematic. While DRC verifies geometric correctness of the layout, LVS verifies electrical correctness.

During LVS verification, the extracted netlist generated from the layout is automatically compared with the original schematic netlist. This comparison confirms whether the designed layout faithfully represents the intended circuit implementation.

The LVS tool checks the following important aspects:

- **Connectivity Matching:** Verifies that all interconnections between devices in the layout correspond exactly to those in the schematic. Any missing connection or unintended short circuit is detected during this step.
- **Device Count Verification:** Ensures that the number of transistors, capacitors, resistors, and other circuit elements present in the layout matches the schematic description.
- **Device Dimension Matching:** Checks whether transistor parameters such as width (W), length (L), number of fingers, and multipliers match those specified in the schematic. Accurate matching of device dimensions is essential for maintaining gain, bandwidth, and matching performance of analog circuits.
- **Pin and Port Consistency:** Verifies that all input/output terminals and bias nodes defined in the schematic are correctly implemented and labeled in the layout.

- **Parameter Consistency:** Ensures that extracted device parameters such as transistor type (NMOS/PMOS), threshold models, and component values correspond to the intended schematic configuration.

LVS verification was performed using the Calibre verification environment integrated with Cadence Virtuoso. The comparison results confirmed that the extracted layout netlist perfectly matches the schematic netlist with no mismatches in connectivity or device parameters. Therefore, the layout is considered **LVS clean** and ready for parasitic extraction (PEX) and post-layout simulation analysis.

2.3 Parasitic Extraction (PEX)

Parasitic Extraction (PEX) is an essential post-layout verification step performed after successful completion of DRC and LVS. The purpose of PEX is to extract layout-dependent parasitic elements such as resistances and capacitances that are introduced due to interconnect routing, device geometries, and physical proximity of layout structures. These parasitic components are not present in schematic-level simulations but significantly affect the actual performance of analog circuits after fabrication.

During the PEX process, the layout is analyzed using the Calibre extraction tool to generate an extracted netlist that includes additional parasitic components associated with:

- **Interconnect Resistance:** Metal routing between circuit nodes introduces distributed resistance which increases signal delay and reduces high-frequency performance. This effect becomes more pronounced in long routing paths and narrow metal lines.
- **Parasitic Capacitance:** Capacitances arise between metal layers, diffusion regions, polysilicon gates, and substrate. These parasitic capacitances increase node loading and directly affect bandwidth, unity gain frequency, and settling time of the operational amplifier.
- **Diffusion and Junction Capacitance:** Source/drain diffusion regions form junction capacitances with the substrate and wells. These capacitances contribute to additional pole formation and reduce overall frequency response.
- **Coupling Capacitance Between Interconnects:** Adjacent routing tracks introduce coupling capacitances which may degrade signal integrity and introduce unwanted feedthrough paths between nodes.

After parasitic extraction, a new extracted view of the layout is generated. This extracted netlist is then used for post-layout simulations to evaluate the realistic performance of the operational amplifier including layout-dependent effects.

The presence of parasitic resistance and capacitance typically leads to degradation in key performance parameters such as:

- Reduction in DC gain due to finite interconnect resistance
- Reduction in unity gain bandwidth due to increased node capacitance
- Reduction in phase margin due to additional parasitic poles

- Degradation of slew rate due to increased effective load capacitance
- Slight variation in common-mode gain and CMRR due to routing asymmetry

Therefore, post-layout simulation after PEX provides a more accurate representation of the practical performance of the two-stage operational amplifier compared to schematic-level simulations.

3 Layout Design Methodology

In analog integrated circuit design, the layout stage plays a crucial role in determining the overall performance of the circuit. Unlike digital circuits, analog circuits are highly sensitive to device mismatches, parasitic capacitances, routing asymmetry, and substrate noise. Therefore, careful layout techniques must be applied to preserve the performance obtained during schematic-level simulations.

While designing the layout of the two-stage operational amplifier, several important analog layout techniques were implemented to ensure proper matching between devices, minimize parasitic effects, and improve overall circuit reliability.

The following layout strategies were used during implementation:

- **Common Centroid Placement for Differential Pair**

The input differential pair transistors were placed using common-centroid geometry. This placement technique minimizes gradient-induced mismatch caused by process variations across the chip such as temperature gradients, oxide thickness variation, and doping concentration variation. Since the differential pair determines the input offset voltage and common-mode rejection ratio (CMRR) of the operational amplifier, accurate matching between these devices is critical.

- **Symmetric Routing of Differential Signals**

Routing for differential signal paths was performed symmetrically to ensure equal parasitic resistance and capacitance along both branches of the differential pair. Symmetric routing helps maintain balanced signal propagation and improves differential gain accuracy while reducing systematic offset errors.

- **Use of Dummy Devices**

Dummy transistors were placed around matched transistor pairs to ensure uniform etching and fabrication conditions across active devices. Dummy devices help reduce edge effects and improve matching accuracy by ensuring that the active devices experience identical processing environments.

- **Minimization of Interconnect Length**

Critical signal paths were routed using minimum possible interconnect length to reduce parasitic resistance and capacitance. This helps in preserving bandwidth, improving unity gain frequency, and maintaining phase margin of the operational amplifier.

- **Guard Ring Implementation**

Guard rings connected to ground were placed around sensitive analog blocks such as the differential pair and current mirrors. Guard rings help isolate the circuit from substrate noise and prevent latch-up by collecting substrate currents generated due to switching activity or external disturbances.

- **Wide Metal Routing for Power Lines**

Wide metal tracks were used for VDD and GND routing to reduce voltage drop due to IR losses and to ensure stable biasing across the circuit. Stable supply routing improves gain stability, reduces noise coupling, and enhances overall reliability of the amplifier.

The implementation of these layout techniques significantly improves device matching and reduces parasitic effects, which directly enhances important performance parameters of the operational amplifier such as:

- Improved Common Mode Rejection Ratio (CMRR)
- Reduced input offset voltage
- Improved DC gain accuracy
- Better phase margin stability
- Higher bandwidth preservation after parasitic extraction
- Enhanced noise immunity and reliability

Thus, careful layout planning ensures that the post-layout performance closely matches the schematic-level simulation results.

4 Layout of Two Stage Opamp (Without Null Resistor)

4.1 Layout

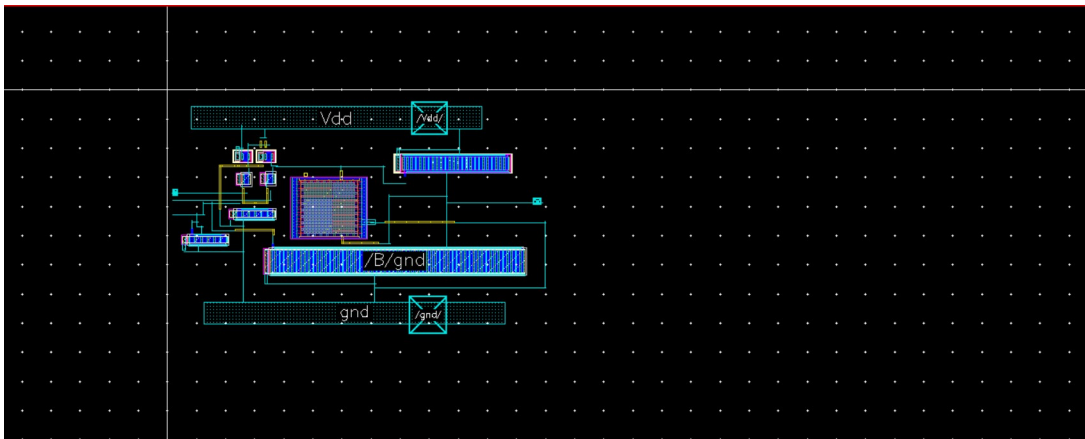


Figure 1: Layout of two-stage opamp without null resistor

4.2 LVS Verification

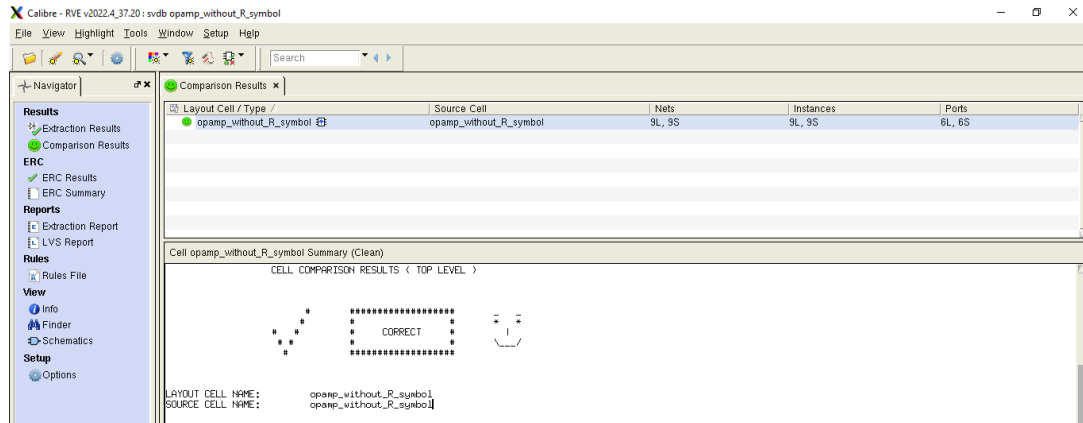


Figure 2: LVS Clean Report

The LVS result confirms:

- device matching correct
- connectivity correct
- layout matches schematic

4.3 Extracted Parasitics

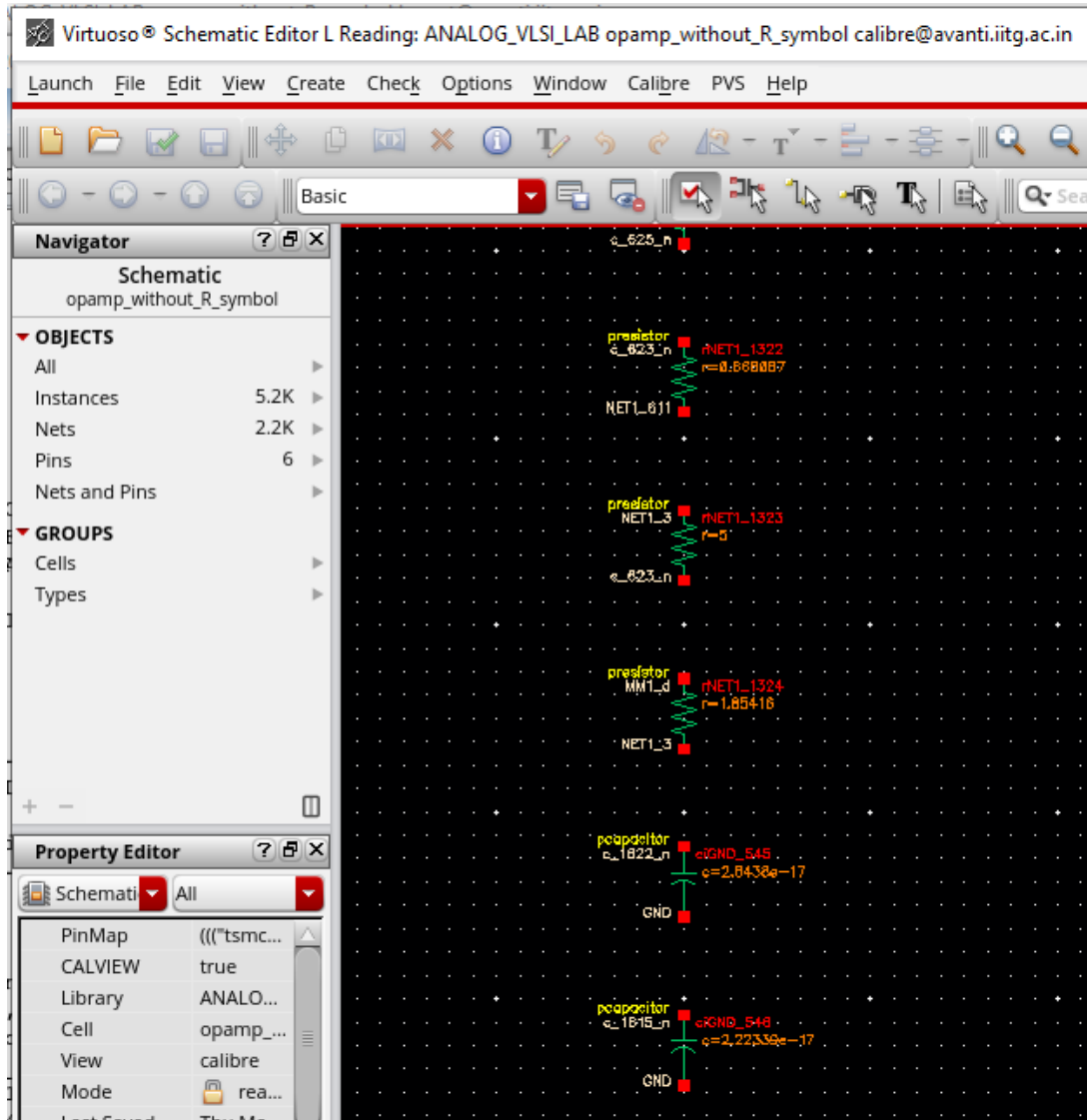


Figure 3: Extracted parasitic resistance and capacitance

5 Pre Layout Simulation Results (Without Null Resistor)

Before performing layout implementation and parasitic extraction, schematic-level simulations were carried out to evaluate the ideal performance of the two-stage operational amplifier without the influence of layout parasitic effects. These simulations provide a reference for comparison with post-layout performance.

The important performance parameters evaluated at schematic level include:

- DC Gain
- Unity Gain Frequency (UGF)
- Phase Margin

- Common Mode Gain
- Slew Rate
- Voltage follower transient response

5.1 Gain and Phase Response Before PEX

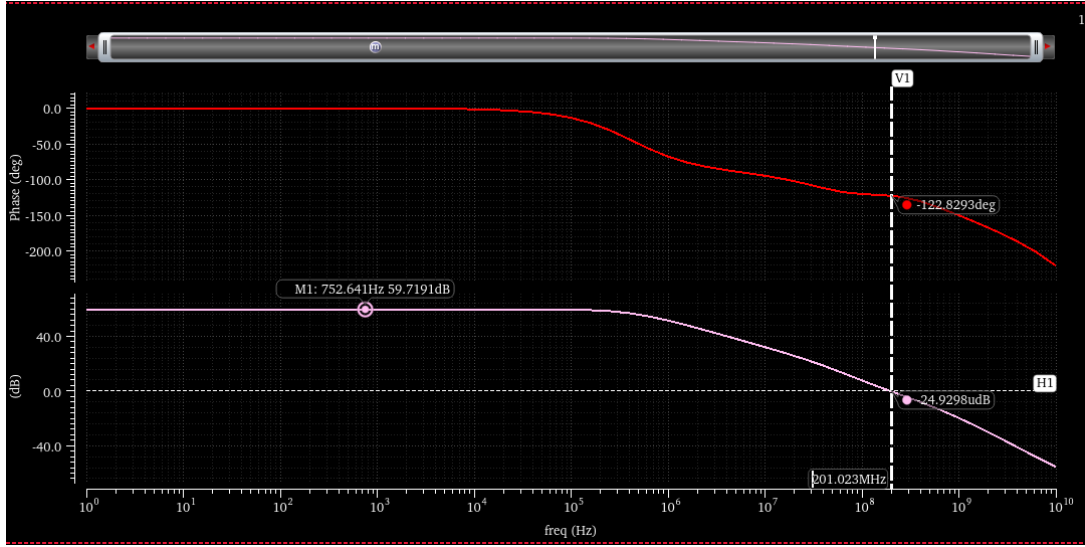


Figure 4: Gain and phase response before parasitic extraction

The AC analysis performed at schematic level shows the frequency response of the operational amplifier. From the plot, the DC gain, unity gain bandwidth, and phase margin of the amplifier were extracted. Since parasitic capacitances are not included at schematic level, the unity gain bandwidth is slightly higher compared to post-layout simulation results.

5.2 Common Mode Gain Before PEX

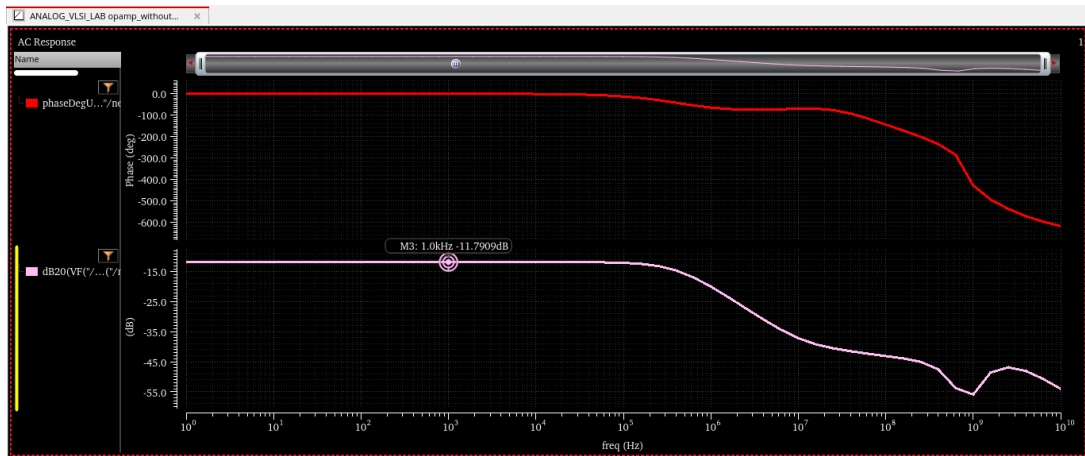


Figure 5: Common mode gain before parasitic extraction

The common-mode gain of the operational amplifier was measured by applying identical input signals to both input terminals. The low value of common-mode gain indicates good matching between the differential pair transistors and proper biasing of the current mirror loads. This ensures high Common Mode Rejection Ratio (CMRR) at schematic level.

5.3 Voltage Follower Transient Response Before PEX

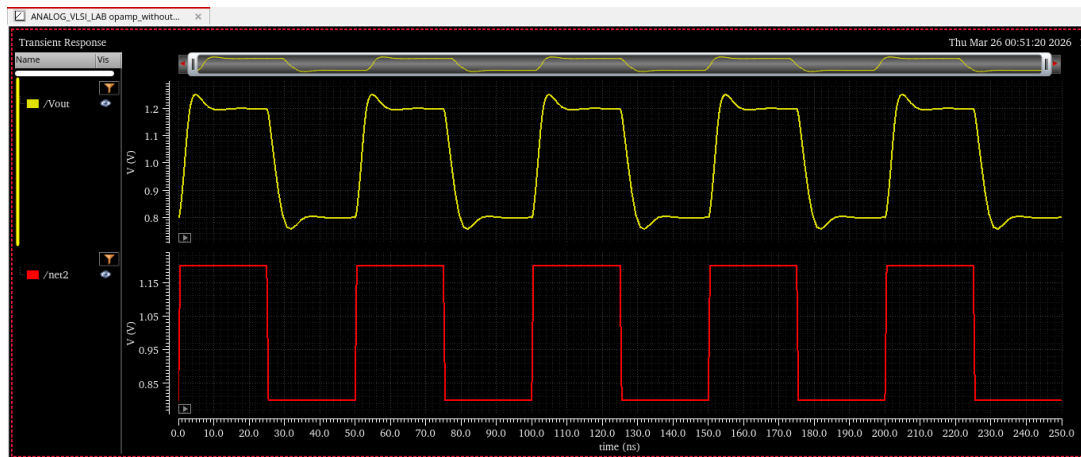


Figure 6: Transient response of voltage follower configuration before parasitic extraction

The operational amplifier was configured as a voltage follower to evaluate its large-signal transient behavior. The output waveform closely follows the input signal, confirming proper closed-loop operation and stability of the amplifier. The transient response also verifies the settling behavior and bandwidth capability of the circuit before layout implementation.

5.4 Slew Rate Before PEX

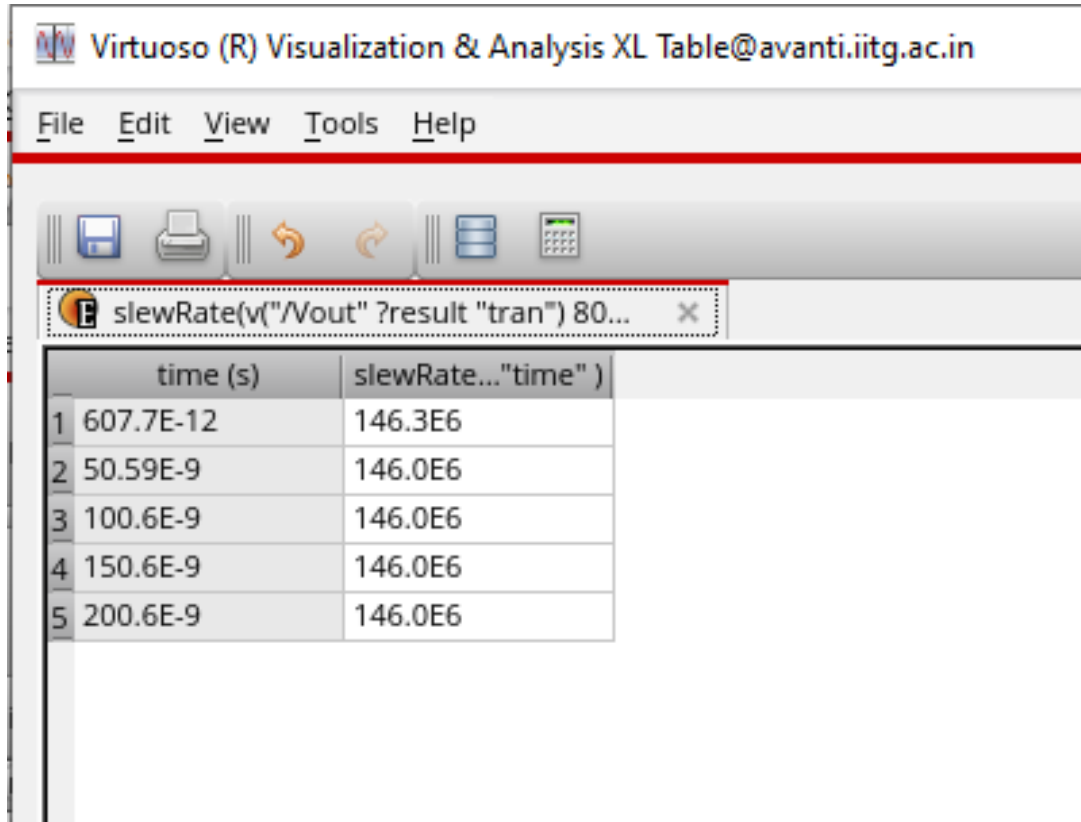


Figure 7: Slew rate measurement before parasitic extraction

The slew rate of the operational amplifier was measured using large-signal transient simulation by applying a step input. The output slope during the rising and falling transitions determines the maximum rate of change of output voltage. Since schematic-level simulations do not include routing parasitic capacitances, the measured slew rate before PEX is higher than the post-layout slew rate.

6 Post Layout Simulation Results (Without Null Resistor)

6.1 Gain and Phase Response After PEX

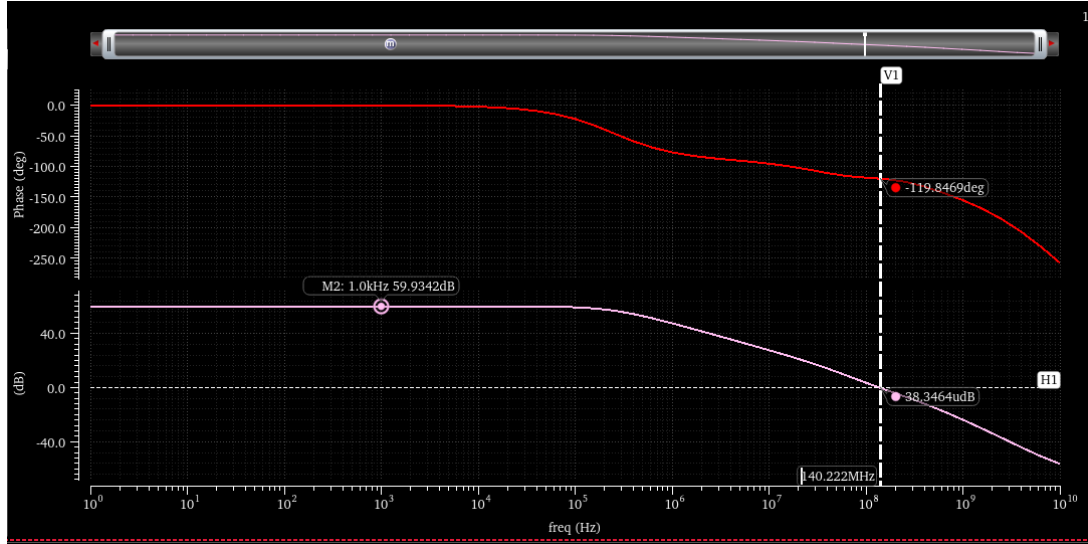


Figure 8: Gain and phase response after PEX

6.2 Common Mode Gain After PEX

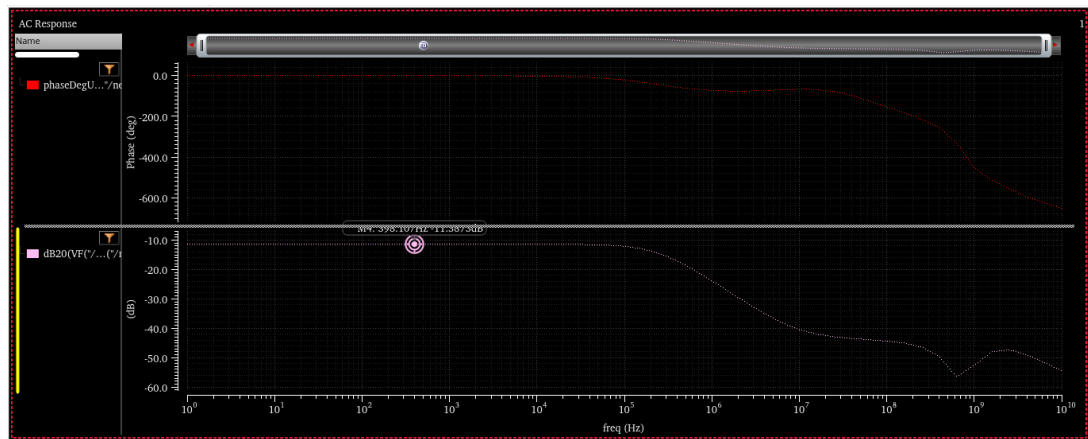


Figure 9: Common mode gain after PEX

6.3 Voltage Follower Transient Response Before PEX

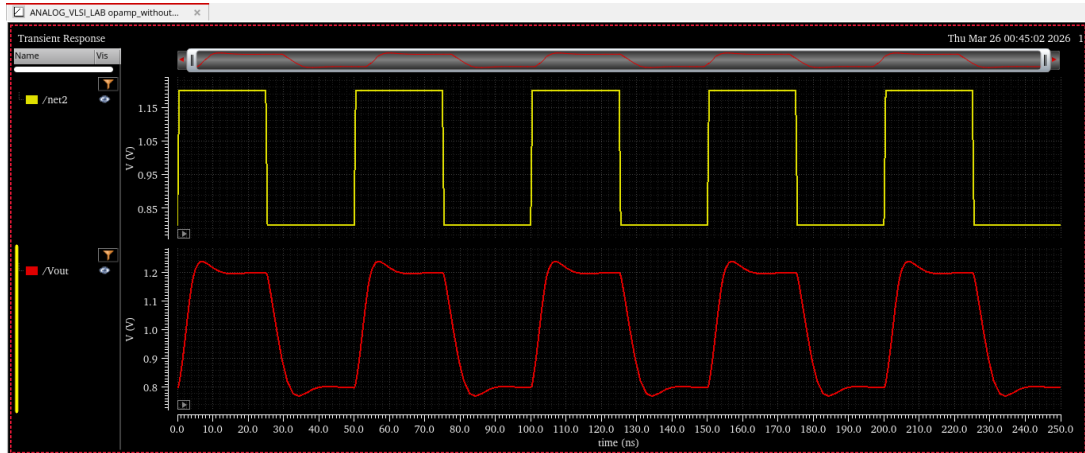


Figure 10: Transient response of voltage follower configuration before parasitic extraction

The operational amplifier was configured as a voltage follower to evaluate its large-signal transient behavior. The output waveform closely follows the input signal, confirming proper closed-loop operation and stability of the amplifier. The transient response also verifies the settling behavior and bandwidth capability of the circuit before layout implementation.

6.4 Slew Rate After PEX

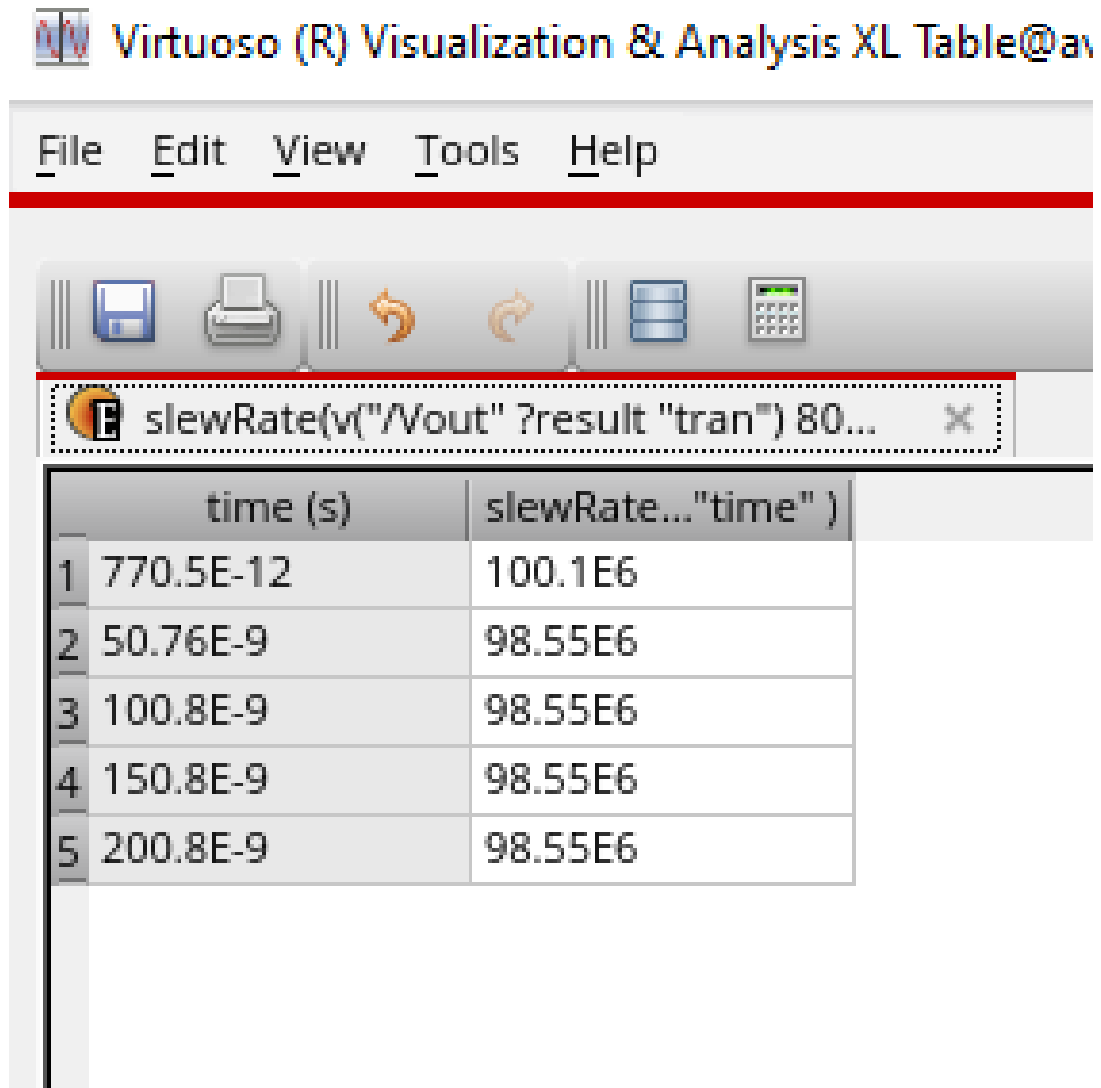


Figure 11: Slew rate after PEX

7 Layout of Two Stage Opamp (With Null Resistor)

7.1 Layout

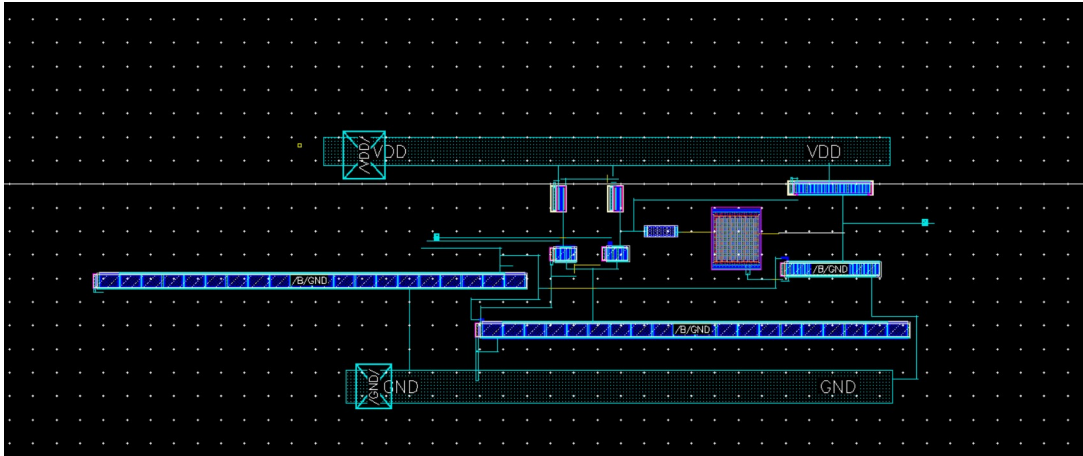


Figure 12: Layout of two-stage opamp with null resistor

7.2 LVS Verification

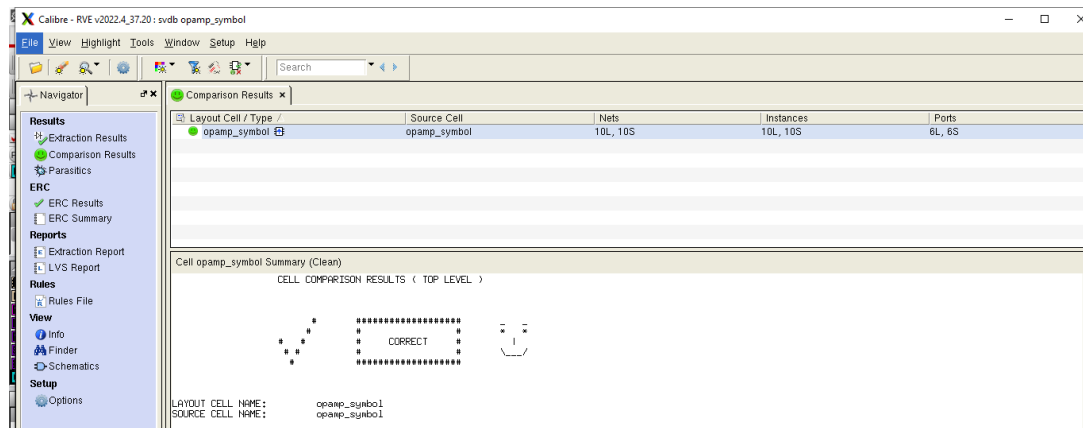


Figure 13: LVS Clean Report

The LVS result confirms:

- device matching correct
- connectivity correct
- layout matches schematic

7.3 Extracted Parasitics

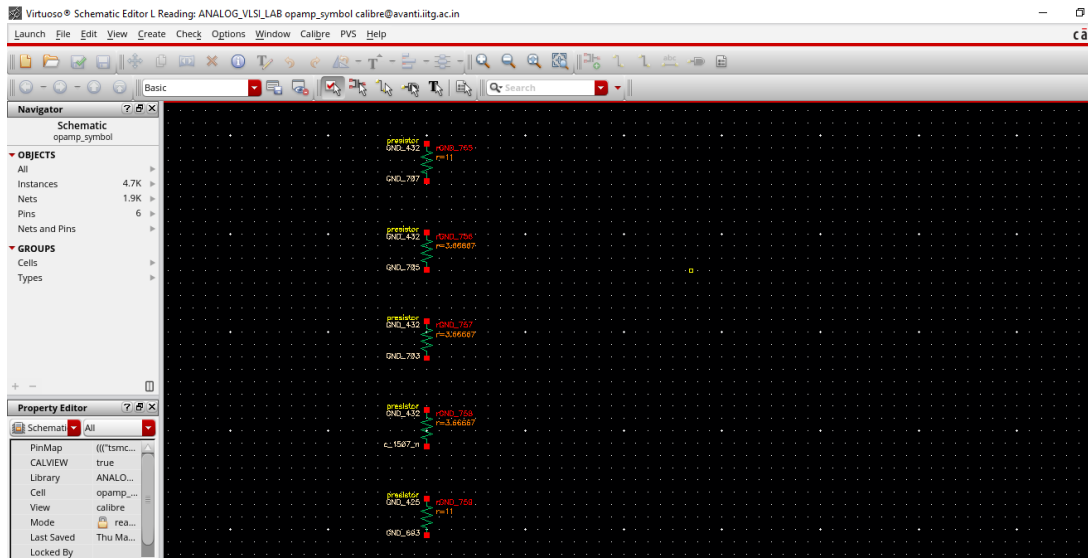


Figure 14: Extracted parasitic resistance and capacitance

8 Pre Layout Simulation Results (With Null Resistor)

Before performing layout implementation and parasitic extraction, schematic-level simulations were carried out to evaluate the ideal performance of the two-stage operational amplifier without the influence of layout parasitic effects. These simulations provide a reference for comparison with post-layout performance.

The important performance parameters evaluated at schematic level include:

- DC Gain
- Unity Gain Frequency (UGF)
- Phase Margin
- Common Mode Gain
- Slew Rate
- Voltage follower transient response

8.1 Gain and Phase Response Before PEX

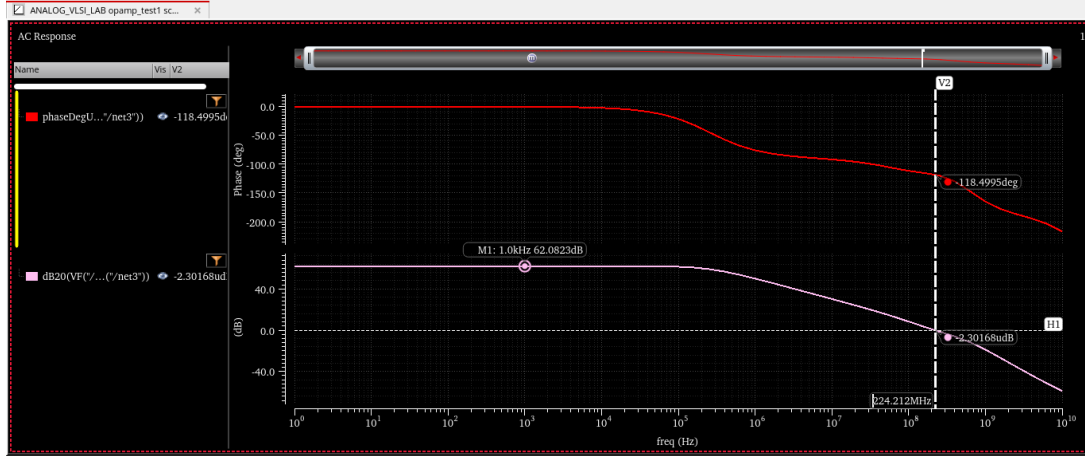


Figure 15: Gain and phase response before parasitic extraction

The AC analysis performed at schematic level shows the frequency response of the operational amplifier. From the plot, the DC gain, unity gain bandwidth, and phase margin of the amplifier were extracted. Since parasitic capacitances are not included at schematic level, the unity gain bandwidth is slightly higher compared to post-layout simulation results.

8.2 Common Mode Gain Before PEX

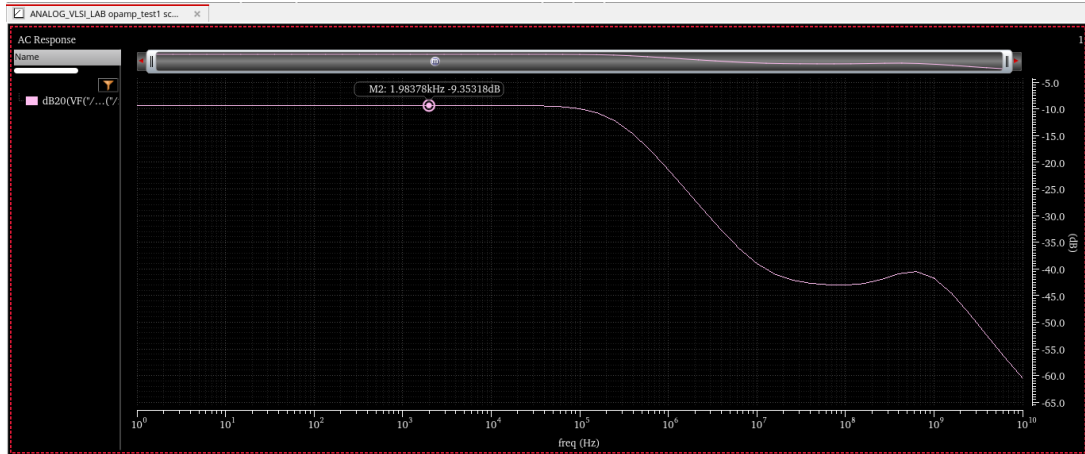


Figure 16: Common mode gain before parasitic extraction

The common-mode gain of the operational amplifier was measured by applying identical input signals to both input terminals. The low value of common-mode gain indicates good matching between the differential pair transistors and proper biasing of the current mirror loads. This ensures high Common Mode Rejection Ratio (CMRR) at schematic level.

8.3 Voltage Follower Transient Response Before PEX

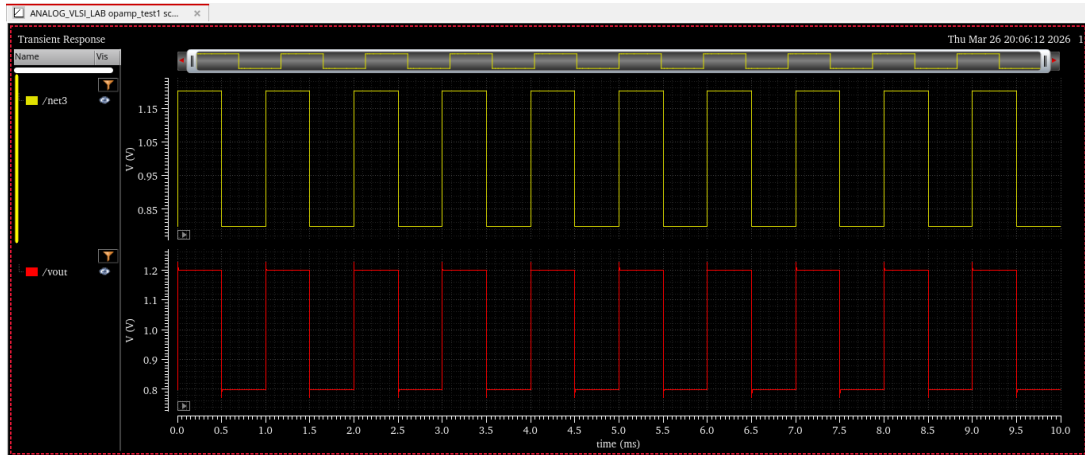


Figure 17: Transient response of voltage follower configuration before parasitic extraction

The operational amplifier was configured as a voltage follower to evaluate its large-signal transient behavior. The output waveform closely follows the input signal, confirming proper closed-loop operation and stability of the amplifier. The transient response also verifies the settling behavior and bandwidth capability of the circuit before layout implementation.

8.4 Slew Rate Before PEX

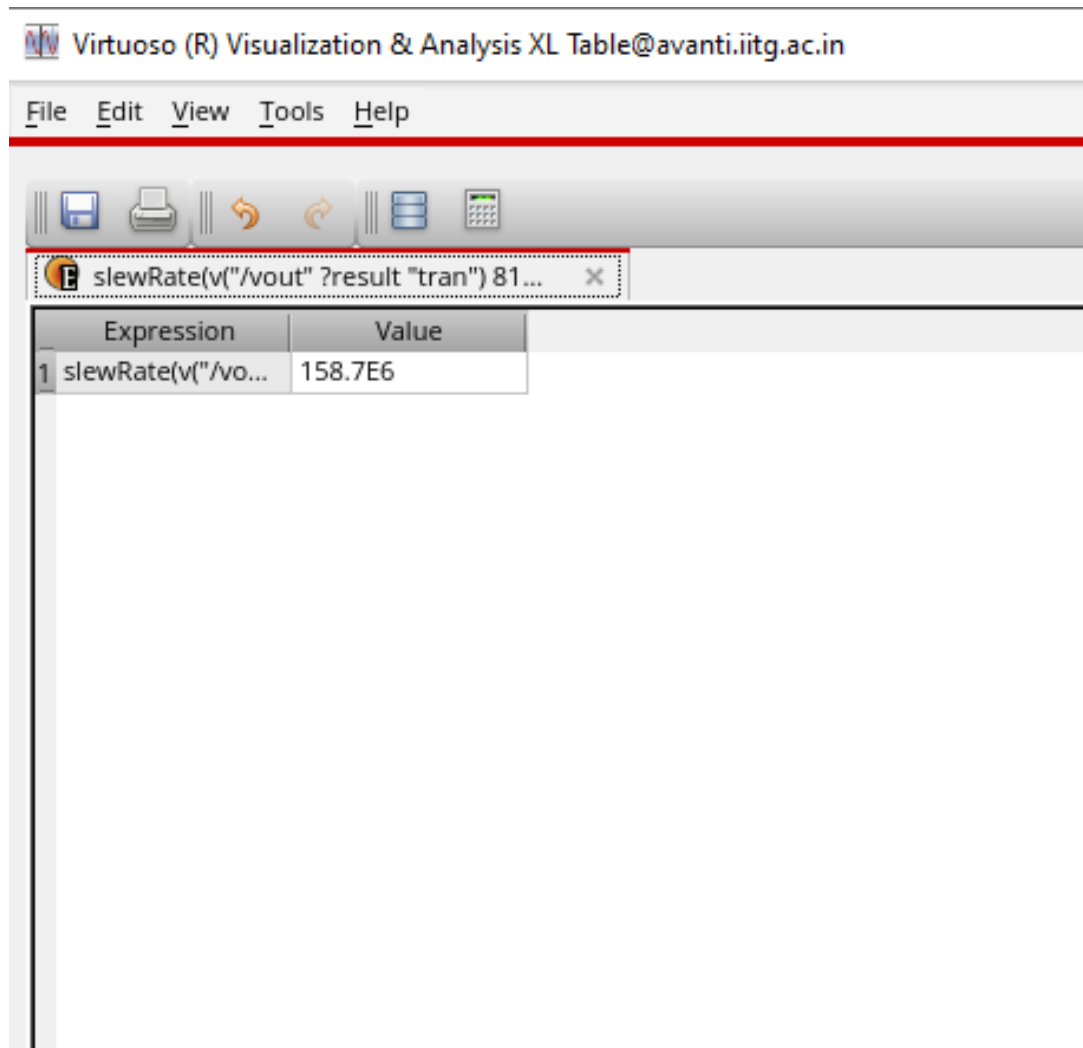


Figure 18: Slew rate measurement before parasitic extraction

The slew rate of the operational amplifier was measured using large-signal transient simulation by applying a step input. The output slope during the rising and falling transitions determines the maximum rate of change of output voltage. Since schematic-level simulations do not include routing parasitic capacitances, the measured slew rate before PEX is higher than the post-layout slew rate.

9 Post Layout Simulation Results (With Null Resistor)

9.1 Gain and Phase Response After PEX

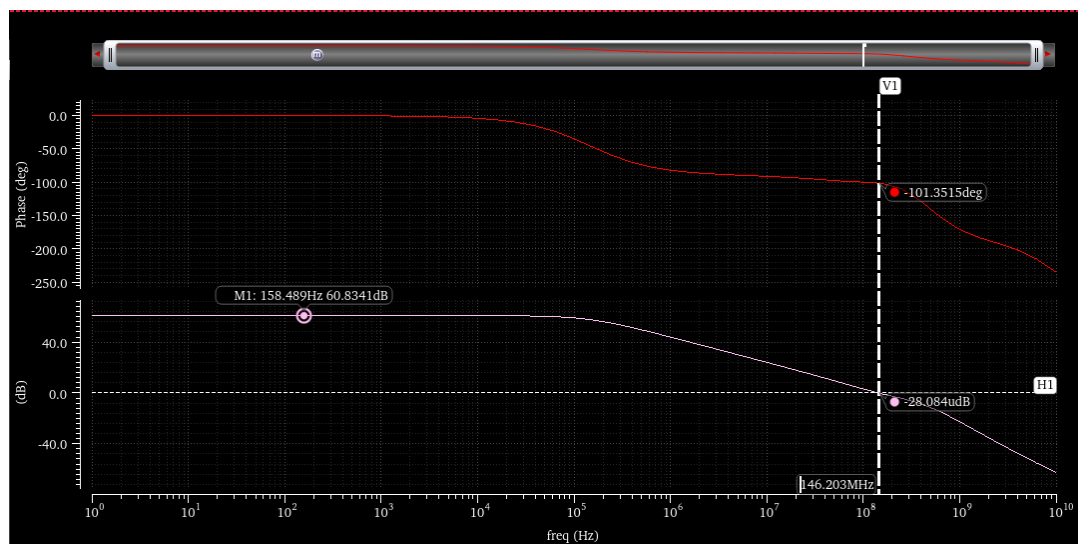


Figure 19: Gain and phase response after PEX

9.2 Common Mode Gain After PEX

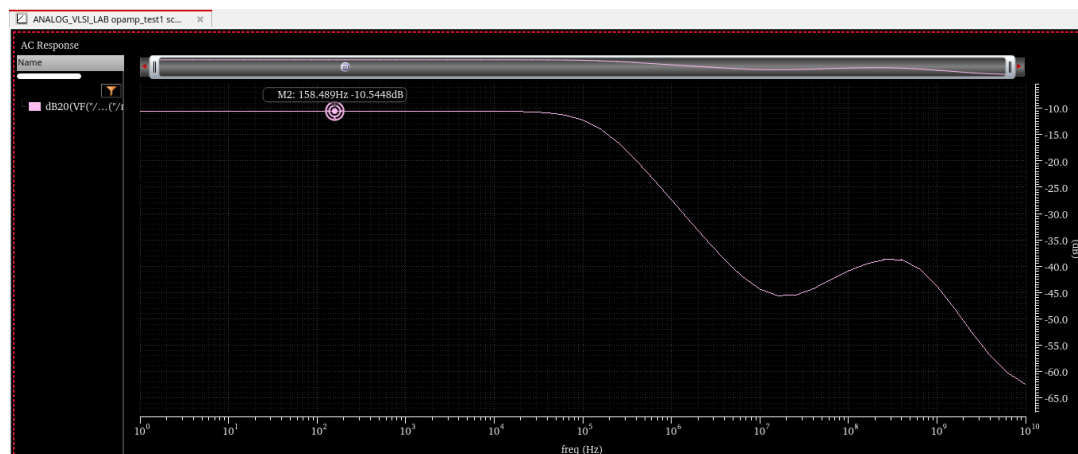


Figure 20: Common mode gain after PEX

9.3 Slew Rate After PEX

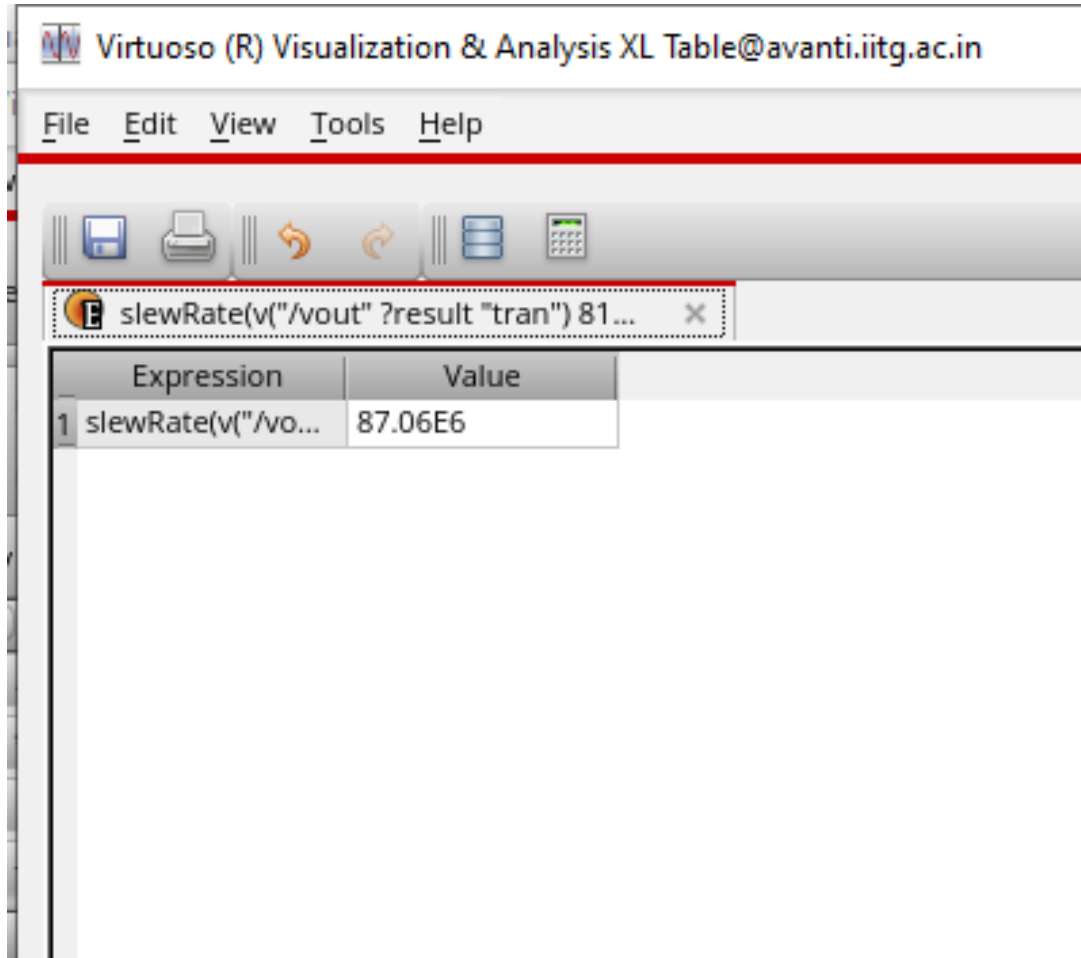


Figure 21: Slew rate after PEX

10 Performance Comparison of Miller Compensated OpAmp Before and After PEX

Parameter	Pre Layout	Post Layout	Variation
DC Gain	59.71 dB	59.9342 dB	slight increase
Unity Gain Frequency	201.023 MHz	140.222 MHz	reduced
Phase Margin	57.1707 deg	60.1531 deg	slightly reduced
Slew Rate	146 V/us	98.55 V/us	reduced
Common Mode Gain	-11.79 dB	-11.3873 dB	nearly same
CMRR	71.5 dB	71.3215 dB	nearly same

Table 1: Performance comparison without null resistor

11 Performance Comparison (With Null Resistor)

Parameter	Pre Layout	Post Layout	Variation
DC Gain	62.0823 dB	60.8341 dB	slight decrease
Unity Gain Frequency	224.212 MHz	146.203 MHz	reduced
Phase Margin	61.5 deg	78.6485 deg	increased
Slew Rate	158 V/us	87.06 V/us	reduced
Common Mode Gain	-9.35318 dB	-10.5448 dB	improved
CMRR	71.43548 dB	71.3829 dB	nearly same

Table 2: Performance comparison without null resistor

12 Discussion

After performing post-layout simulations using the extracted netlist, a slight degradation in performance parameters was observed compared to schematic-level simulations. This degradation is expected in analog integrated circuits because layout implementation introduces unavoidable parasitic resistance and capacitance that are not present in ideal schematic simulations.

The primary sources responsible for performance variation between pre-layout and post-layout results include:

- **Routing Resistance**

Metal interconnects introduce distributed resistance along signal paths. This additional resistance reduces effective node voltage swing at high frequencies and slightly lowers the overall voltage gain of the operational amplifier.

- **Routing Capacitance**

Parasitic capacitance between metal layers and neighboring routing tracks increases the effective load capacitance at internal nodes. This shifts pole locations toward lower frequencies, resulting in reduction of unity gain bandwidth and phase margin.

- **Diffusion Capacitance**

Source and drain diffusion regions form junction capacitances with substrate and wells. These capacitances introduce additional poles in the small-signal response of the amplifier, which affects frequency response and stability.

- **Contact and Via Resistance**

Contacts between metal layers and diffusion regions introduce additional series resistance. These resistances slightly degrade signal propagation speed and reduce slew rate performance.

- **Device Mismatch Due to Layout Effects**

Even with symmetric placement techniques, small mismatches occur due to process gradients and routing asymmetry. These mismatches affect common-mode gain and slightly modify the Common Mode Rejection Ratio (CMRR).

It was observed from post-layout simulations that the dominant performance variations occurred in:

- reduction in DC gain due to increased effective output resistance loading
- reduction in unity gain frequency due to additional parasitic capacitances
- slight variation in phase margin due to pole shifting
- reduction in slew rate because of increased effective node capacitance
- minor variation in common-mode gain and CMRR due to layout asymmetry

Despite these variations, the overall performance of both versions of the two-stage operational amplifier (with and without null resistor) remained close to the schematic-level design targets. The use of careful layout techniques such as common-centroid placement, symmetric routing, guard rings, and minimized interconnect lengths helped reduce the impact of parasitic effects.

Therefore, the post-layout simulation results confirm that the designed layouts are robust and suitable for practical implementation while maintaining acceptable performance accuracy compared to pre-layout simulations.

13 Conclusion

In this experiment, the layout of a two-stage Miller compensated operational amplifier was successfully implemented using Cadence Virtuoso in 0.18 μm CMOS technology. Two versions of the operational amplifier were analyzed:

- Two-stage op-amp without null resistor
- Two-stage op-amp with null resistor

After completing the layout design, standard physical verification steps were performed to ensure correctness of implementation. The following verification results were obtained:

- The layout satisfied all fabrication constraints and was verified to be **DRC clean**
- Electrical equivalence between schematic and layout was confirmed through **LVS matching**
- Layout-dependent parasitic resistances and capacitances were extracted successfully using **PEX**
- Post-layout simulations were performed using the extracted netlist to evaluate realistic circuit performance

Comparison between schematic-level and post-layout simulation results showed a slight degradation in important performance parameters such as DC gain, unity gain bandwidth, phase margin, and slew rate. These variations are expected due to the presence of routing parasitics, diffusion capacitances, and contact resistances introduced during layout implementation.

However, the performance degradation remained within acceptable limits and the amplifier continued to satisfy the required design specifications. The use of proper analog layout techniques such as common-centroid placement, symmetric routing, guard rings, and minimized interconnect lengths helped preserve circuit performance after parasitic extraction.

Furthermore, comparison between the two compensation approaches demonstrated that the introduction of a null resistor improves stability and allows better control over phase margin with reduced sensitivity to parasitic effects.

Therefore, the layout implementation and post-layout verification confirm that the designed two-stage operational amplifier operates reliably under practical layout conditions and is suitable for physical realization in CMOS technology.